

# RW BLE iOS SDK User Guide

## 1. Introduction

This document explains the functional APIs and usage scenarios provided by the SDK.

This document applies only to RW company Bluetooth devices.

### 1.1 Supported Platforms and Languages

- iOS 12 and above, Objective-C language.

### 1.2 Terminology

- App: Refers to the application running on a mobile phone or tablet.
- Device: Refers to wearable hardware devices, such as watches and rings.
- Upload: Refers to data sent from the device to the App.
- Download: Refers to data sent from the App to the device.

### 1.3 Notes

1. This document uses Objective-C for all examples. If you use Swift, you must import the corresponding Objective-C header files in the project's Bridging Header.
2. The SDK does not provide a simulator version, because BLE cannot be debugged on the simulator, and some third-party libraries used by the SDK do not support the simulator environment.

## 2. Quick Start

### Step 1: Install the latest version of Xcode

To develop with the RW BLE iOS SDK, Xcode must be installed.

### Step 2: Manually add dependencies

Manually add `DHBleSDK.framework` to your project.

#### Frameworks, Libraries, and Embedded Content

| Name                                                                                                       | Embed        |        |
|------------------------------------------------------------------------------------------------------------|--------------|--------|
|  DHBleSDK.framework     | Do Not Embed | ⬆️⬇️⬆️ |
|  DHFoundation.framework | Embed & Sign | ⬆️⬇️⬆️ |
|  DHUIKit.framework      | Embed & Sign | ⬆️⬇️⬆️ |

### Step 3: Configure info.plist

```
//Add Bluetooth usage descriptions to info.plist.
NSBluetoothAlwaysUsageDescription
NSBluetoothPeripheralUsageDescription
```

### Step 4: Initialize the SDK

```
//Initialize the DHBleSDK in AppDelegate.
- (void)initBleSDK{
    [DHBleCentralManager setLogStatus:YES];
    [DHBleCentralManager initWithServiceUids:@[]];
}
```

#### ⚠ Caution

When logging is enabled `[DHBleCentralManager setLogStatus:YES]` , log files will be stored in the `Document/DeviceLog` directory.

## 3. API Reference)

### 3.1 Device Scanning, Connection, Binding, and Reconnection

#### 3.1.1 Scan for Devices

Description: Call `startScan` to scan for BLE devices and implement the `DHBleConnectDelegate`.

If the returned `DHPeripheralModel` has an empty `macAddr`, it indicates that the device is already paired in system settings.

```
// 1. 开始搜索
[DHBleCentralManager startScan];

//2. 设备委托
[DHBleCentralManager sharedInstance].connectDelegate = self;

//3. DHBleConnectDelegate接口会回调搜索到的蓝牙设备
- (void)centralManagerDidDiscoverPeripheral:(NSArray <DHPeripheralModel *>*)peripherals
```

#### 3.1.2 Stop Scanning

Description: Stop scanning for BLE devices.

```
[DHBleCentralManager stopScan];
```

#### 3.1.3 Connect Device

Description: Connect to a specified device and set the connection delegate.

```
// 1. Initialize and register the callback.
[DHBleCentralManager sharedInstance].connectDelegate = self;

// 2. Connect device
DHPeripheralModel *deviceModel = self.deviceArray[indexPath.row];
[DHBleCentralManager connectDeviceWithModel:deviceModel];

// 3. Implement and receive callbacks for Bluetooth connection status.
@protocol DHBleConnectDelegate <NSObject>
```

`DHBleConnectDelegate` Interface Description:

| method                                | illustrate                                                                                                                                      |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| centralManagerDidDiscoverPeripheral   | After calling <code>startScan</code> , the system will return a callback when a device is found.                                                |
| centralManagerDidConnectPeripheral    | After calling <code>connectDeviceWithModel</code> , the function will return upon successful connection.                                        |
| centralManagerDidFunctionMenu         | The function will return after successfully obtaining the device configuration table; business operations should be performed after this point. |
| centralManagerDidDisconnectPeripheral | Called when Bluetooth disconnects. Implement the new method with <code>reason</code> to obtain the disconnection reason.                        |
| centralManagerDidFailedPeripheral     | Bluetooth failure will trigger a callback.                                                                                                      |
| centralManagerDidUpdateState          | The Bluetooth switch state change will trigger a callback.                                                                                      |

Disconnection callback with reason:

```
-(void)centralManagerDidDisconnectPeripheral:(CBPeripheral *)peripheral reason:
(DHBluetoothReason)reason
```

| Enum Value                                       | Description                           |
|--------------------------------------------------|---------------------------------------|
| <code>DHBluetoothReasonUnknown</code>            | Unknown or normal disconnection       |
| <code>DHBluetoothReasonManualDisconnect</code>   | Disconnected by the App               |
| <code>DHBluetoothReasonPasswordAuthFailed</code> | Device password authentication failed |

If the new method with `reason` is implemented, the SDK will not also call the original `centralManagerDidDisconnectPeripheral:` method. If it is not implemented, the original callback remains compatible.

#### Tip

After connecting, business operations should only be performed after the `centralManagerDidFunctionMenu` method has been called.

(1) Listening to `BluetoothNotificationConnectStateChange` can also provide information about changes in the connection state.

```
[[NSNotificationCenter defaultCenter] addObserver:self selector:@selector(connectStateChange:)
name:BluetoothNotificationConnectStateChange object:nil];

- (void)connectStateChange:(NSNotification *)ntf
{
    NSLog(@"NewHomeController connectStateChange");
    if ([DHBluetoothManager sharedInstance].isConnected){
        self.infoDeviceStateLb.text = @"Connected";
    }
    else{
        self.infoDeviceStateLb.text = @"Disconnected";
    }
}
```

(2) `[DHBluetoothManager isConnected]` can be used to check if the connection was successful.

### 3.1.4 Disconnect the device.

Interface description: Disconnect the currently connected device;

```
[DHBluetoothManager disconnectDevice];
```

### 3.1.5 Binding and automatic reconnection, unbinding

#### 3.1.5.1 DHBluetoothManager Save to the current device locally.

After setting this, the current device's UDID will be saved, and it will automatically reconnect.

Method Description:

```
+(void)setBindedStatus:(BOOL)isBinded
```

Example of usage:

```
//Save locally, and reopening the file will reconnect.
[DHBluetoothManager setBindedStatus:YES];
```

#### 3.1.5.2 DHBluetoothManager Delete locally saved data

After this setting is applied, the UDID of the current device will be deleted, similar to unlinking the device locally.

Method Description:

```
+(void)setBindedStatus:(BOOL)isBinded
```

Example of usage:

```
[DHBluetoothManager setBindedStatus:NO];
[DHBluetoothManager disconnectDevice];
```

### 3.1.6 Equipment Configuration Table

#### Important

Due to the variety of device models and their differing supported features, a feature table has been introduced to allow users to check the supported functions of each device. Please refer to the DeviceFuncV2Model class for details. The feature table content can be saved according to your specific business needs.

```
-(void)centralManagerDidFunctionMenu:(DeviceFuncV2Model *)deviceFuncModel
```

DeviceFuncV2Model class attribute definitions:

| attribute                       | illustrate                                                                         |
|---------------------------------|------------------------------------------------------------------------------------|
| isPushMsgEnableSwitch           | Enable or disable message control switch                                           |
| pushMsgSwitchValue              | Supported message types, low 32 bits (bit0-bit31)                                  |
| pushMsgSwitchValue2             | Supported message types, high 32 bits (bit32-bit63); defaults to 0 on old devices  |
| activityDataInterval            | Today's step-detail interval in minutes; an unconfigured value is normalized to 60 |
| isAlarm                         | Does it support an alarm clock?                                                    |
| isBackLight                     | Does it support screen sleep time settings?                                        |
| isSupportWorkout3               | Does it support multiple sports modes?                                             |
| isSupportMuslimCountSwitch      | Does it support enabling/disabling Muslim prayers?                                 |
| isSupportHrSpO2Alert            | Does it support HR and SpO2 alarm notification functions?                          |
| isSupportMotoVibrationLevel     | Does it support motor vibration alerts?                                            |
| isSupportAlarmVibrationDuration | Does it support alarm vibration duration setting?                                  |
| isSupportVibrationInterval      | Does it support vibration interval setting?                                        |
| isDataTypeActivity              | Does it support step counting?                                                     |
| isDataTypeHeart                 | Does it support heart rate monitoring?                                             |
| isDataTypeBloodPressure         | Does it support blood pressure measurement?                                        |
| isDataTypeSleep                 | Does it support sleep mode?                                                        |
| isDataTypeSPO2                  | Does it support blood oxygen monitoring?                                           |
| isDataTypeHRV                   | Does it support heart rate variability?                                            |
| isDataTypeStress                | Does it support pressure sensitivity?                                              |
| isDataTypeBloodSugar            | Does it support blood glucose monitoring?                                          |
| isDataTypeMuslimCount           | Do you support giving compliments/praise?                                          |
| isSupportMuslimTimeDisplayMode  | Does it support Muslim time display mode?                                          |
| isSupportSensorRawPPG           | Does it support PPG Green raw data?                                                |
| isSupportPPGMonitoring          | Does it support PPG timed monitoring?                                              |
| isSupportTemperatureMonitoring  | Does it support temperature timed monitoring?                                      |
| isSupportCountReminder          | Does it support count reminder interval setting?                                   |
| isSupportSensorRawACC           | Does it support ACC raw data?                                                      |
| isSupportSensorRawPPGRed        | Does it support PPG Red raw data?                                                  |
|                                 |                                                                                    |

|                             |                                                 |
|-----------------------------|-------------------------------------------------|
| isSupportSensorRawIR        | Does it support IR (infrared) raw data?         |
| isSupportSensorRawSleep     | Does it support sleep real-time data?           |
| isSupportFallDetect         | Does it support fall detection alert?           |
| isSupportRecording          | Does it support recording function?             |
| isSupportDevicePasswordAuth | Does it support device password authentication? |
| isSupportScreenControl      | Does it support instant screen on/off control?  |

### 3.1.8 Use an External CBCentralManager for Scanning and Let the SDK Connect

Description: If the customer App integrates multiple BLE SDKs and uses its own unified BLE scanning entry, pass the scanning `CBCentralManager` to this SDK. The customer App is only responsible for scanning. Connection, service discovery, characteristic discovery, data transfer, and disconnection are still handled by the SDK.

#### Important

A `CBPeripheral` is bound to the `CBCentralManager` that discovered it. The `peripheral` passed to the SDK must come from the currently set `externalCentralManager`. The SDK does not use private APIs to verify the source. If the Central and Peripheral do not match, the connection may fail or time out.

Method Description:

```
// Initialize the SDK and specify an external CBCentralManager.
// Pass nil to use the SDK-managed mode.
+ (void)initWithServiceUuids:(NSArray <NSString *>)uuids
    externalCentralManager:(nullable CBCentralManager *)centralManager;

// Set an external CBCentralManager at runtime.
// It can only be set when the SDK is not connecting or connected.
+ (void)setExternalCentralManager:(nullable CBCentralManager *)centralManager;

// Check whether the SDK is using an external CBCentralManager.
+ (BOOL)isUsingExternalCentralManager;

// Connect to a device discovered by the customer's unified scanner.
+ (void)connectPeripheral:(CBPeripheral *)peripheral
    advertisementData:(nullable NSDictionary *)advertisementData
    RSSI:(nullable NSNumber *)RSSI;
```

Parameter Description:

| parameter         | type             | description                                                                                                                                                                                                       |
|-------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| centralManager    | CBCentralManager | The same Central used by the customer App for unified scanning. The SDK will use it to connect.                                                                                                                   |
| peripheral        | CBPeripheral     | The device object discovered by <code>centralManager</code> . The connection must use this object.                                                                                                                |
| advertisementData | NSDictionary     | Advertisement data used to complete <code>DHPeripheralModel</code> fields such as <code>macAddr</code> , <code>deviceModel</code> , and device name. It is not required for the connection itself and can be nil. |
| RSSI              | NSNumber         | Signal strength used to complete <code>DHPeripheralModel.rssi</code> . It is not required for the connection itself and can be nil.                                                                               |

Example:

```
// 1. The customer App scans devices with its own CBCentralManager.
// centralManager, peripheral, advertisementData, and RSSI come from the customer's scan callback.

// 2. Pass the same CBCentralManager to the SDK.
[DHBLECentralManager initWithServiceUuids:@[]
                    externalCentralManager:centralManager];

// Or set the external Central at runtime if the SDK has already been initialized.
[DHBLECentralManager setExternalCentralManager:centralManager];

// 3. Set the SDK connection delegate.
[DHBLECentralManager sharedInstance].connectDelegate = self;

// 4. Let the SDK start the connection.
[DHBLECentralManager connectPeripheral:peripheral
                    advertisementData:advertisementData
                        RSSI:RSSI];
```

Notes:

- The old integration method `[DHBLECentralManager initWithServiceUuids:@[]]` remains unchanged, and the SDK will create its own `CBCentralManager`.
- In external Central mode, the SDK connects by using the same `CBCentralManager` passed by the customer App.
- After the SDK starts connecting, it takes over `centralManager.delegate`, and subsequent connection callbacks are handled by the SDK.
- The customer App should stop its own scanning flow before handing the device to the SDK for connection.
- Do not switch `externalCentralManager` while connecting, connected, discovering services, or synchronizing data.
- For automatic reconnection in external Central mode, the customer App must keep the passed `centralManager` instance alive.

## 3.2 Device function operation

Error code definitions for API calls `SendStateCode`:

| SendStateCode           | value | illustrate                         |
|-------------------------|-------|------------------------------------|
| SendState_OK            | 0     | OK                                 |
| SendState_BLENoOpen     | 1     | Failure, Bluetooth is not enabled. |
| SendState_BLEDisconnect | 2     | Failure, Bluetooth not connected.  |
| SendState_TimeOut       | 3     | Failure, Timeout                   |
| SendState_Failed        | 4     | fail                               |
| SendState_DuplicateSend | 5     | Repeat command                     |
| SendState_NotSupport    | 6     | Device not supported               |

### 3.2.1 Basic function command interface

#### 3.2.1.0 Get SDK Version

Get the SDK version number.

Method Description:

[DHBLECommand getSDKVersion]

Example of usage:

```
NSLog(@"%@", [DHBLECommand getSDKVersion]);
```

#### 3.2.1.1 Get Bluetooth MAC address

Because the iOS system cannot obtain the MAC address from the broadcast packet after pairing, this method is provided to retrieve it.

Method Description:

+ (void)ringGetMacAddress:(void(^)(int code, id data))block

Return parameter description:

| parameter | type              | illustrate |                |
|-----------|-------------------|------------|----------------|
| model     | DHDeviceInfoModel | class      | macAddr: Mac地址 |

Example of usage:

```
[DHBLECommand ringGetMacAddress:^(int code, id _Nonnull data) {
    DHDeviceInfoModel *tDeviceInfoData = data;
    NSLog(@"mac: %@", tDeviceInfoData.macAddr);
}];
```

#### 3.2.1.2 Set user information

User information settings are related to step count, calories burned, and distance. During device initialization, the default settings are: gender 1, age 18, height 170cm, and weight 65 kg.

Method Description:

```
+ (void)setUserInfo:(DHUserInfoSetModel *)model block:(void(^)(int code, id data))block
```

Parameter Description:

| parameter | type               |       | illustrate                                                                                                                                                                                                      |
|-----------|--------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model     | DHUserInfoSetModel | class | gender: Gender (0. Female, 1. Male)<br>Height: height in cm, floating-point number<br>Weight: weight in kg, floating-point number<br>Step Goal: step count target value, currently no functionality<br>Age: age |

Example of usage:

```
DHUserInfoSetModel *userInfoModel = [[DHUserInfoSetModel alloc] init];
userInfoModel.gender = 1;
userInfoModel.height = 170;
userInfoModel.weight = 600;
userInfoModel.stepGoal = 8000;
userInfoModel.age = 20;
[DHbleCommand setUserInfo:userInfoModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"set ok");
    }
}];
```

### 3.2.1.3 Get Device Information

Retrieves the device model, firmware version, and UI version.

Method:

```
+ (void)getFirmwareVersion:(void(^)(int code, id data))block
```

Return value:

| DHFirmwareVersionModel property | Type     | Description                                                |
|---------------------------------|----------|------------------------------------------------------------|
| deviceModel                     | NSString | Device model, the unique identifier for each product model |
| firmwareVersion                 | NSString | Firmware version                                           |
| uiVersion                       | NSString | UI version                                                 |

**Important:** Before an OTA update, verify that the device `deviceModel` matches the target device model of the firmware file. Start the update only when they match. Do not update a device with firmware for a different model.

Example:

```
[DHBleCommand getFirmwareVersion:^(int code, id _Nonnull data) {
    if (code == 0 && [data isKindOfClass:[DHFirmwareVersionModel class]]) {
        DHFirmwareVersionModel *model = data;
        NSLog(@"model %@ firmware %@ UI %@",
            model.deviceModel, model.firmwareVersion, model.uiVersion);
    }
}];
```

### 3.2.1.4 Get Battery Level

Retrieves the current device battery level.

Method:

```
+ (void)getBattery:(void(^)(int code, id data))block
```

Return value:

| DHBatteryInfoModel property | Type      | Description                          |
|-----------------------------|-----------|--------------------------------------|
| battery                     | NSInteger | Remaining battery level, range 0-100 |

Example:

```
[DHBleCommand getBattery:^(int code, id _Nonnull data) {
    if (code == 0 && [data isKindOfClass:[DHBatteryInfoModel class]]) {
        DHBatteryInfoModel *model = data;
        NSLog(@"battery %zd", model.battery);
    }
}];
```

### 3.2.1.5 Get and Set the Video Control Mode

Set the ring gesture control mode; [this function requires pairing with Bluetooth HID.](#)

Parameter Description:

| Parameter | Type | Description        | Value                               |
|-----------|------|--------------------|-------------------------------------|
| isOpen    | int  | Video control mode | 0: Off, 1: Video, 2: Book, 3: Music |

```
//Set video control mode
DHVideoHidSetModel *tModeSetModel = [[DHVideoHidSetModel alloc] init];
tModeSetModel.isOpen = 1; //0 Off, 1 Video, 2 Book, 3 Music
[DHBleCommand setVideoHid:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setVideoHid OK");
    }
}];

// Get video control mode
[DHBleCommand getVideoHid:^(int code, id _Nonnull data) {
    if (code == 0){
        DHVideoHidSetModel *model = data;
        NSLog(@"getVideoHid OK mode %d", model.isOpen);
    }
}];
```

3.2.1.6 Get and set LED screen brightness.

Method Description:

```
+(void)getRingLEDLight:(void(^)(int code, id data))block;

+(void)setRingLEDLight:(DHLedLightSetModel *)model block:(void(^)(int code, id data))block;
```

Parameter Description:

| parameter | type               |       | illustrate                                                                                                       |
|-----------|--------------------|-------|------------------------------------------------------------------------------------------------------------------|
| model     | DHLedLightSetModel | class | isOpen: false means off, true means (Levels 1-3)<br>Light Level: 1 (dim light), 2 (soft light), 3 (bright light) |

Example of usage:

```
// Get the LED screen brightness level.
[DHBluetooth getRingLEDLight:^(int code, id _Nonnull data) {
    if (code == 0){
        DHLedLightSetModel *model = data;
        NSLog(@"getRingLEDLight OK 开关 %d", model.isOpen);
    }
}];

// Set the LED screen brightness.
DHLedLightSetModel *tModeSetModel = [[DHLedLightSetModel alloc] init];
tModeSetModel.isOpen = YES; //
tModeSetModel.lightLevel = 3; //1 (dim light), 2 (soft light), 3 (bright light)
[DHBluetooth setRingLEDLight:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setRingLEDLight OK");
    }
}];
```

3.2.1.7 Get and set the wearing position.

Gets or sets the ring wearing position.  
Configuration table property: `isWearDir`.

Methods:

```
+ (void)getRingWearHand:(void(^)(int code, id data))block

+ (void)setRingWearHand:(UInt8)wearHand block:(void(^)(int code, id data))block
```

Parameter:

| Parameter | Type  | Description      | Values                      |
|-----------|-------|------------------|-----------------------------|
| wearHand  | UInt8 | Wearing position | 0: Left hand; 1: Right hand |

Return value:

| Return data | Type     | Description                                         |
|-------------|----------|-----------------------------------------------------|
| data        | NSNumber | Wearing position: 0 for left hand, 1 for right hand |

Example:

```
// Get the wearing position
[DHBleCommand getRingWearHand:^(int code, id _Nonnull data) {
    if (code == 0){
        NSInteger wearHand = [data integerValue]; // 0: Left; 1: Right
        NSLog(@"wearing position %zd", wearHand);
    }
}];

// Set the wearing position
UInt8 wearHand = 0; // 0: Left; 1: Right
[DHBleCommand setRingWearHand:wearHand block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setRingWearHand OK");
    }
}];
```

### 3.2.1.8 Starting and stopping photo taking

Enable camera control when the APP enters its custom camera page. The device can then notify the APP to take a photo by gesture. Disable camera control when the APP leaves the camera page.

Configuration table property: `isTakePhoto`.

Receive device photo events through `BluetoothNotificationCameraTakePicture`.

Method:

```
+ (void)controlCamera:(NSInteger)type block:(void(^)(int code, id data))block
```

Parameter:

| Parameter | Type      | Description    | Values                |
|-----------|-----------|----------------|-----------------------|
| type      | NSInteger | Camera control | 0: Disable; 1: Enable |

Example:

```
// Call when the APP enters the camera page
- (void)openCameraPage {
    [[NSNotificationCenter defaultCenter] addObserver:self
                                              selector:@selector(cameraTakePictureNotification:)
                                              name:BluetoothNotificationCameraTakePicture
                                              object:nil];
    [DHBleCommand controlCamera:1 block:^(int code, id _Nonnull data) {
        NSLog(@"enable camera control code=%d", code);
    }];
}

// Call when the APP leaves the camera page
- (void)closeCameraPage {
    [DHBleCommand controlCamera:0 block:^(int code, id _Nonnull data) {
        NSLog(@"disable camera control code=%d", code);
    }];
}
```

```

    }];
    [[NSNotificationCenter defaultCenter] removeObserver:self
                                              name:BluetoothNotificationCameraTakePicture
                                              object:nil];
}

- (void)cameraTakePictureNotification:(NSNotification *)notification {
    // Take a photo with the APP's custom camera here
}

```

### 3.2.1.9 Find devices

After initiating the search function, the device's light or screen will turn on.

Method Description:

```
+(void)controlFindDeviceBegin:(void(^)(int code, id data))block;
```

Example of usage:

```

[DHBluetooth controlFindDeviceBegin:^(int code, id _Nonnull data) {

}
];

```

### 3.2.1.10 Turn off the device, restore factory settings.

Method Description:

```
+(void)controlDevice:(NSInteger)type block:(void(^)(int code, id data))block;
```

Parameter Description:

| parameter | type      |    | illustrate                        |
|-----------|-----------|----|-----------------------------------|
| type      | NSInteger | 整形 | 1: Turn off<br>2: restore factory |

Example of usage:

```

// 1关机 2 恢复出厂
[DHBluetooth controlDevice:1 block:^(int code, id _Nonnull data) {

}
];

```

### 3.2.1.11 Alarm

Configuration table attribute: `isAlarm`

#### 3.2.1.11.1 Get the alarms that have been set.

Method Description:

```
+(void)getAlarms:(void(^)(int code, id data))block
```

Example of usage:

```
//获取设备里已保存的闹钟
[DHBleCommand getAlarms:^(int code, id _Nonnull data) {
    if (code == 0) {
        NSArray *tAlarmList = data;
        NSLog(@"getAlarms %zd", tAlarmList.count);
    }
}];
```

### 3.2.1.11.2 Set an alarm

The current protocol does not support modifying individual alarms. Any operation to switch on/off or delete a single alarm requires resending the entire alarm configuration.

Method Description:

```
+(void)setAlarms:(NSArray <DHAlarmSetModel *>*)alarms block:(void(^)(int code, id data))block;
```

Parameter Description:

| parameter | type | illustrate | illustrate                                                                                                                                                             |
|-----------|------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alarms    | List | List       | isOpen: true Open/false Off<br>repeats: IntArray(7) Sunday to Saturday, set the corresponding element to 1 for repetition.<br>hour: start hour<br>minute: start minute |

Example of usage:

```
//设置闹钟
DHAlarmSetModel *tAlarm1 = [[DHAlarmSetModel alloc] init];
tAlarm1.hour = 07; //时
tAlarm1.minute = 00; //分
tAlarm1.isOpen = true; //开关
tAlarm1.repeats = @[0,0,0,0,0,0,1]; //重复周期,周六重复

DHAlarmSetModel *tAlarm2 = [[DHAlarmSetModel alloc] init];
tAlarm2.hour = 8;
tAlarm2.minute = 00;
tAlarm2.isOpen = true;
tAlarm2.repeats = @[]; //单次闹钟

[DHBleCommand setAlarms:@[tAlarm1, tAlarm2] block:^(int code, id _Nonnull data) {
}];
```

### 3.2.1.11.3 Delete all alarms

By passing an empty array to the `setAlarms` parameter, you can delete all alarms.

```
[DHBleCommand setAlarms:@[] block:^(int code, id _Nonnull data) {
}];
```

### 3.2.1.12 Setting and retrieving the number of vibrations

Set the number of times the device vibrates;

Configuration table property: `isSupportMotoVibrationLevel`

Method Description:

```
+(void)setRingMotorLevel:(NSInteger)motorLevel motorNum:(NSInteger)motorNum block:(void(^)(int code, id data))block;

+(void)getRingMotorLevel:(void(^)(int code, id data))block;
```

Parameter Description:

| parameter  | type | illustrate | illustrate                                                                                                                  |
|------------|------|------------|-----------------------------------------------------------------------------------------------------------------------------|
| motorLevel | Int  | 整形         | Vibration intensity: 0: Off, 1: Low, 2: Medium, 3: High; <i>This function is not defined and can be ignored</i>             |
| motorNum   | Int  | 整形         | The number of vibrations can be set (0-6 times), with a default setting of 2 times. Setting it to 0 will disable vibration. |

Example of usage:

```
//设置
[DHBleCommand setRingMotorLevel:1 motorNum:2 block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"set ok");
    }
}];

//获取
[DHBleCommand getRingMotorLevel:^(int code, id _Nonnull data) {
    if (code == 0){
        DHVibrationLevelModel *tVibrationModel = data;
        NSLog(@"getRingMotorLevel Level %d num %d", tVibrationModel.vibrationLevel, tVibrationModel.vibrationNumber);
    }
}];
```

3.2.1.13 Setting and retrieving screen sleep mode settings.

Set screen sleep mode and time;

Configuration table property: `isBackLightSleepMode`

Method Description:

```
+(void)setDisplaySleepMode:(DHBrightTimeSetModel *)model block:(void(^)(int code, id data))block

+(void)getDisplaySleepMode:(void(^)(int code, id data))block
```

Parameter Description:

| parameter            | type  | illustrate | illustrate                                                                                                                                                                           |
|----------------------|-------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHBrightTimeSetModel | class |            | sleepOpen: Switch on (YES) or off (NO)<br>sleepStartHour: Start time (hour)<br>sleepStartMin: Start time (minute)<br>sleepEndHour: End time (hour)<br>sleepEndMin: End time (minute) |

Example of usage:

```
//设置
DHBrightTimeSetModel *sleepModel = [[DHBrightTimeSetModel alloc] init];
sleepModel.sleepOpen = YES;
sleepModel.sleepStartHour = 20;
sleepModel.sleepStartMin = 00;
sleepModel.sleepEndHour = 06;
sleepModel.sleepEndMin = 00;
[DHBleCommand setDisplaySleepMode:sleepModel block:^(int code, id _Nonnull data) {

}];

//获取
[DHBleCommand getDisplaySleepMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHBrightTimeSetModel *tModel = data;
        NSLog(@"getDisplaySleepMode sleepOpen %d sleepStartHour %d sleepStartMin %d", tModel.sleepOpen,
tModel.sleepStartHour, tModel.sleepEndMin);
    }
}];
```

3.2.1.14 Setting and retrieving notification push settings

`DeviceFuncV2Model.isPushMsgEnableSwitch` indicates whether the device supports message-push switch settings. `pushMsgSwitchValue` and `pushMsgSwitchValue2` are capability masks for bit0-bit31 and bit32-bit63; they are not the device's current switch states.

When opening the message-push settings page, call `ringGetAncs:` to fetch the current switches from the device. On success, `data` is a `DHAncsSetModel`. Modify that model and call `ringSetAncs:block:` to write the switches back to the device.

| Purpose                       | Property/API                                                       | Result                                      |
|-------------------------------|--------------------------------------------------------------------|---------------------------------------------|
| Check feature support         | <code>DeviceFuncV2Model.isPushMsgEnableSwitch</code>               | Whether message-push switches are supported |
| Check supported message types | <code>pushMsgSwitchValue</code> / <code>pushMsgSwitchValue2</code> | Supported message types, bit0-bit63         |
| Fetch current switches        | <code>ringGetAncs:</code>                                          | <code>DHAncsSetModel</code>                 |
| Set current switches          | <code>ringSetAncs:block:</code>                                    | Setting result                              |

Method Description:

```
+(void)ringSetAncs:(DHAncsSetModel *)model block:(void(^)(int code, id data))block

+(void)ringGetAncs:(void(^)(int code, id data))block
```

Parameter Description:

| parameter      | type  | illustrate | illustrate                                      |
|----------------|-------|------------|-------------------------------------------------|
| DHAncsSetModel | class |            | See the definition of the DHAncsSetModel class. |

Example of usage:

```
//设置
DHAncsSetModel *tAncsModel = [[DHAncsSetModel alloc] init];
tAncsModel.isSMS = YES;
[DHBleCommand ringSetAncs:tAncsModel block:^(int code, id _Nonnull data) {

}

}];

//获取
[DHBleCommand ringGetAncs:^(int code, id _Nonnull data) {
    if (code == 0){
        DHAncsSetModel *ancsModel = data;
        NSLog(@"SMS=%d Wechat=%d", ancsModel.isSMS, ancsModel.isWechat);
    }
}];

//按下面的顺序确定当前设备支持哪些应用消息。
UInt8 tBitRow = i;
if (i == 24){ //其它
    tBitRow = 0;
}
if (i > 0 && (tMsgSwitchValue & (1 << tBitRow)) < 1){ //不支持的消息全设置为false,不管从设备读到的值。
    continue;
}

self.msgOpenFlagArr = [NSMutableArray arrayWithArray:@[
    @(self.model.isCall),
    @(self.model.isSMS),
    @(self.model.isEmail),
    @(self.model.isSkype),
    @(self.model.isFacebook),
    @(self.model.isWhatsapp),
    @(self.model.isLine),
    @(self.model.isInstagram),
    @(self.model.isKakaotalk),
    @(self.model.isGmail),
    @(self.model.isTwitter),
    @(self.model.isLinkedin),
    @(self.model.isJLSinaWeiBo),

    @(self.model.isQQ),
    @(self.model.isWechat),
    @(self.model.isJLBand),
    @(self.model.isJLTelegram),
    @(self.model.isJLBetween),
```

```

    @(self.model.isJLNavercafe),
    @(self.model.isYoutube),
    @(self.model.isJLNetflix), //(1<<21)
    @(self.model.isMax), // (1<<22)
    @(self.model.isVkim), // (1<<23)
    //          @(self.model.isMessenger),

    @(self.model.isOther)
  ]];

```

### 3.2.1.15 Get and set whether the "likes" feature is enabled.

Set whether the "like" function is enabled;

Configuration table property: `isSupportMuslimCountSwitch`

Method Description:

```
+(void)setMuslimCountSwitch:(UInt8)isOpen block:(void(^)(int code, id data))block
```

```
+(void)getMuslimCountSwitch:(void(^)(int code, id data))block
```

Parameter Description:

| parameter | type | illustrate | illustrate        |
|-----------|------|------------|-------------------|
| isOpen    | Int  |            | 0: Off<br>1: Open |

Example of usage:

```

//设置
[DHBluetooth setMuslimCountSwitch:1 block:^(int code, id _Nonnull data) {

}];

//获取
[DHBluetooth getMuslimCountSwitch:^(int code, id _Nonnull data) {
    if (code == 0){
        Boolean tOpen = [data boolValue];
        NSLog(@"getMuslimCountSwitch tOpen %d", tOpen);
    }
}];

```

### 3.2.1.16 Get and set heart rate/blood oxygen alarm configuration.

This function allows you to set heart rate and blood oxygen level notification and alarm data; the alarm notification will be sent via `BluetoothNotificationRingHealthOverAlert`.

Configuration table attribute: `isSupportHrSpO2Alert`

Method Description:

```
+(void)getHRAAlert:(void(^)(int code, id data))block
```

```
+(void)setHRAAlert:(DHHRAAlertModel *)overModel block:(void(^)(int code, id data))block
```

```
+(void)getSP02Alert:(void(^)(int code, id data))block
```

```
+(void)setSP02Alert:(DHHRAAlertModel *)overModel block:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter       | type  | illustrate | illustrate                                                                                                                                                                                                                                                                                              |
|-----------------|-------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHHRAAlertModel | class |            | isOpen: YES (On), NO (Off);<br>overValue: Alarm threshold, default values are heart rate exceeding 160 and blood oxygen below 94%.<br>underValue: An alarm will be triggered if the value is below the set threshold; if the retrieved value is 0xff, it indicates that this function is not supported. |

**Note:** If the `underValue` returned by `getHRAAlert()` is 0xff, it means this feature is not supported.

Example of usage:

```
//set
DHHRAAlertModel *tHRAAlertModel = [[DHHRAAlertModel alloc] init];
tHRAAlertModel.isOpen = YES;
tHRAAlertModel.overValue = 160;
tHRAAlertModel.underValue = 0xff;
[DHBleCommand setHRAAlert:tHRAAlertModel block:^(int code, id _Nonnull data) {

}];

//get
[DHBleCommand getHRAAlert:^(int code, id _Nonnull data) {
    if (code == 0){
        DHHRAAlertModel *model = data;
        NSLog(@"getHRAAlert %d %zd", model.isOpen, model.overValue);
    }
}];

// Alarm notification push
[[NSNotificationCenter defaultCenter] addObserver:self selector:@selector(healthOverAlert:)
name:BluetoothNotificationRingHealthOverAlert object:nil];

- (void)healthOverAlert:(NSNotification *)ntf
{
    NSDictionary *tUserInfo = ntf.userInfo;
    NSInteger tType = [tUserInfo[@"type"] integerValue];
    NSInteger tValue = [tUserInfo[@"value"] integerValue];

    if (tType == 0){ //HeartRate Alert

    }
    else if (tType == 1){ //SP02

    }

}
```

### 3.2.1.17 Get and set screen timeout duration.

Configuration table property: `isBackLight`

Method Description:

```
+(void)getBrightTime:(void(^)(int code, id data))block;
```

```
+(void)setBrightTime:(DHBrightTimeSetModel *)model block:(void(^)(int code, id data))block;
```

Parameter Description:

| DHBrightTimeSetModel Parameter | Type   | Description                                                                 | Value |
|--------------------------------|--------|-----------------------------------------------------------------------------|-------|
| duration                       | Int    | Screen-on duration, in seconds (s), range 0-30s;                            |       |
| durationNums                   | String | Supported duration values by the device; if available, separated by commas; |       |

### 3.2.1.18 Get and Set Wrist-Raise Screen-On Duration

Configuration table property: `isSupportRaisescreen`

Method Description:

```
+(void)ringGetGesture:(void(^)(int code, id data))block;
```

```
+(void)ringSetGesture:(DHGestureSetModel *)model block:(void(^)(int code, id data))block;
```

Parameter Description:

| BrightScreenBean Parameter | Type | Description                    | Value |
|----------------------------|------|--------------------------------|-------|
| isOpen                     | Int  | true: enabled; false: disabled |       |
| startHour                  | Int  | Start time (hour)              |       |
| startMin                   | Int  | Start time (minute)            |       |
| endHour                    | Int  | End time (hour)                |       |
| endMin                     | Int  | End time (minute)              |       |

### 3.2.1.19 Set Time Format (12-Hour / 24-Hour)

This setting only applies to devices with a display.

Method Description:

```
+(void)ringSetTimeformat:(UInt8)timeformat block:(void(^)(int code, id data))block;
```

Parameter Description:

| Parameter  | Type | Description              |  |
|------------|------|--------------------------|--|
| timeformat | Int  | 0: 24-Hour<br>1: 12-Hour |  |

3.2.1.20 Alarm Vibration Duration Setting and Getting

Set the alarm vibration count;  
Configuration table property: `isSupportAlarmVibrationDuration`

Method Description:

```
+(void)setAlarmVibrationDuration:(UInt8)count block:(void(^)(int code, id data))block  
  
+(void)getAlarmVibrationDuration:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type  | Description | Value                                                  |
|-----------|-------|-------------|--------------------------------------------------------|
| count     | UInt8 | Integer     | Vibration count (0-6), default 2, 0 means no vibration |

Example of usage:

```
//Set  
[DHBleCommand setAlarmVibrationDuration:2 block:^(int code, id _Nonnull data) {  
    if (code == 0){  
        NSLog(@"setAlarmVibrationDuration OK");  
    }  
}];  
  
//Get  
[DHBleCommand getAlarmVibrationDuration:^(int code, id _Nonnull data) {  
    if (code == 0){  
        NSLog(@"getAlarmVibrationDuration count %@", data);  
    }  
}];
```

3.2.1.21 Touch Event Notification

Device touch event notification, actively reported by the device. Touch operations are reported regardless of screen state. The APP defines the response behavior.

Received via `BluetoothNotificationTouchEvent` notification.

**Note:** This is a device-side customization. Before using it, confirm that the device manufacturer has integrated and enabled it in the firmware. If it has not been customized or enabled, the APP will not receive touch event notifications.

Notification userInfo data:

| Field     | Description | Value                                                                                                                     |
|-----------|-------------|---------------------------------------------------------------------------------------------------------------------------|
| keyType   | Key type    | 1: Touch key (default), 2: Fall (requires fall detect enabled 3.2.1.24)                                                   |
| touchType | Touch type  | 1: Single tap, 2: Double tap, 3: Triple tap, 4: Long press, 5: Flick.<br>When key type=2 (fall), touch type defaults to 1 |

Example of usage:

```
[[NSNotificationCenter defaultCenter] addObserver:self selector:@selector(touchEventNotification:)
name:BluetoothNotificationTouchEvent object:nil];

- (void)touchEventNotification:(NSNotification *)ntf
{
    NSDictionary *tUserInfo = ntf.userInfo;
    NSInteger keyType = [tUserInfo[@"keyType"] integerValue]; // 1: Touch key
    NSInteger touchType = [tUserInfo[@"touchType"] integerValue]; // 1: Single tap, 2: Double tap, 3:
Triple tap, 4: Long press, 5: Flick
    NSLog(@"TouchEvent keyType=%zd touchType=%zd", keyType, touchType);
}
```

### 3.2.1.22 Vibration Interval Setting and Getting

Set the interval time between each vibration, used to adjust vibration rhythm;

Configuration table property: `isSupportVibrationInterval`

Method Description:

```
+(void)setVibrationInterval:(UInt16)intervalMs block:(void(^)(int code, id data))block
```

```
+(void)getVibrationInterval:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter  | Type   | Description | Value                                         |
|------------|--------|-------------|-----------------------------------------------|
| intervalMs | UInt16 | Integer     | Interval duration (100-1000ms), default 500ms |

Example of usage:

```
//Set
[DHBluetooth setVibrationInterval:500 block:^(int code, id _Nonnull data) {
    if (code == 0){ NSLog(@"setVibrationInterval OK"); }
}];

//Get
[DHBluetooth getVibrationInterval:^(int code, id _Nonnull data) {
    if (code == 0){ NSLog(@"getVibrationInterval %@ms", data); }
}];
```

### 3.2.1.23 HR Calibration (Factory Test)

Start device heart rate calibration mode. After sending the calibration command, the device returns 2 responses:

1st response: result=0 (calibrating); 2nd response: result≠0 (calibration done).

block callback data is NSDictionary: `testMode` (UInt8) + `result` (UInt32, 0=calibrating, non-0=done).

Method Description:

```
+(void)startFactoryTest:(UInt8)testMode block:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type  | Description | Value                |
|-----------|-------|-------------|----------------------|
| testMode  | UInt8 | Test mode   | 0x15: HR Calibration |

Example of usage:

```
[DHBleCommand startFactoryTest:0x15 block:^(int code, id _Nonnull data) {
    if (code == 0 && [data isKindOfClass:[NSDictionary class]]){
        NSDictionary *info = data;
        NSInteger result = [info[@"result"] integerValue];
        if (result == 0){
            NSLog(@"HR calibrating...");
        } else {
            NSLog(@"HR calibration done, result=%zd", result);
        }
    }
}];
```

### 3.2.1.24 Fall Detection Setting

Set or get the fall detection alert switch. When enabled, the device will report fall events via touch event notification (3.2.1.21).

Fall events are reported through `BluetoothNotificationRingTouchEvent` notification, with keyType=2 indicating a fall event.

Configuration table property: `isSupportFallDetect`

Method Description:

```
+(void)setFallDetect:(UInt8)enable block:(void(^)(int code, id data))block
+(void)getFallDetect:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type  | Description | Value         |
|-----------|-------|-------------|---------------|
| enable    | UInt8 | Switch      | 0: off, 1: on |

Example of usage:

```
//Get fall detect switch
[DHBleCommand getFallDetect:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"getFallDetect OK: %@", data);
    }
}];

//Set fall detect on
[DHBleCommand setFallDetect:1 block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setFallDetect ON OK");
    }
}];
```

### 3.2.1.25 Count Reminder Interval Setting

Set or get the count reminder interval. When enabled, after the user completes a count operation, the device starts timing and vibrates once when the interval is reached to remind the user to continue counting.

Configuration table property: `isSupportCountReminder`

Method Description:

```
+(void)setCountReminderInterval:(UInt8)interval block:(void(^)(int code, id data))block
```

```
+(void)getCountReminderInterval:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type  | Description      | Value                                   |
|-----------|-------|------------------|-----------------------------------------|
| interval  | UInt8 | Interval minutes | 0: off, 30/60/90/120: reminder interval |

Example of usage:

```
//Get count reminder interval
[DHBluetooth getCountReminderInterval:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"CountReminderInterval: %@ min", data);
    }
}];

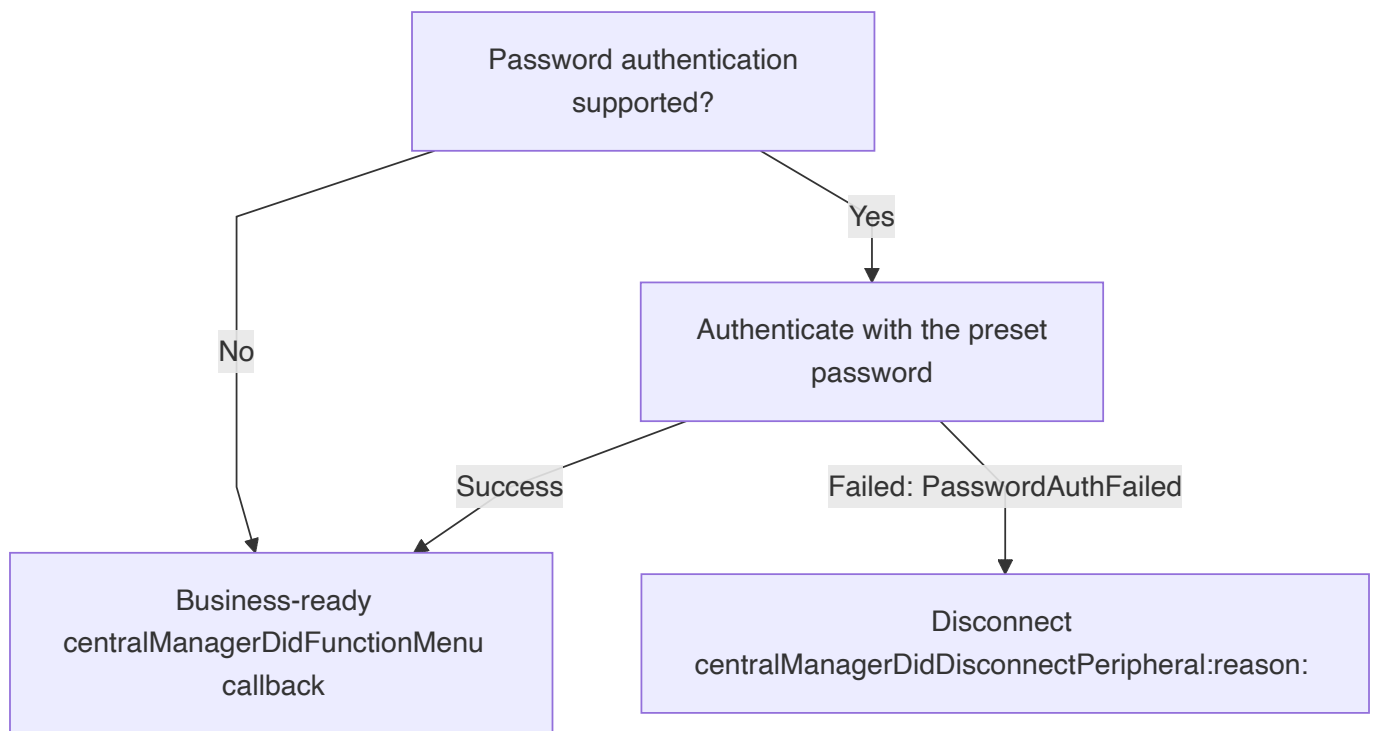
//Set count reminder interval to 60 minutes
[DHBluetooth setCountReminderInterval:60 block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setCountReminderInterval OK");
    }
}];
```

### 3.2.1.26 Device Password Authentication

Check `isSupportDevicePasswordAuth` in the device configuration table to determine whether the device supports password authentication.

The password must contain four digits. A `nil` or empty value is treated as the default password `0000`.

For a supported device, `centralManagerDidFunctionMenu` is called only after authentication succeeds. If authentication fails, the SDK disconnects and returns `DHBluetoothDisconnectReasonPasswordAuthFailed`. Unsupported devices continue to use the original connection flow.



### 3.2.1.26.1 Set the Automatic Authentication Password

```
+(void)prepareAutoPassword:(nullable NSString *)password
```

Set the password used by the SDK for automatic authentication. It may be configured after SDK initialization, but it must be called before connecting. The same value is also used when reconnecting to a locally bound device.

Input Parameter:

| Parameter | Type     | Description                                                                              |
|-----------|----------|------------------------------------------------------------------------------------------|
| password  | NSString | Four-digit password; <code>nil</code> or an empty string is treated as <code>0000</code> |

Callback Result:

| Callback Method                               | Result                                  | Description                                            |
|-----------------------------------------------|-----------------------------------------|--------------------------------------------------------|
| centralManagerDidFunctionMenu:peripheral:     | DeviceFuncV2Model                       | Authentication succeeded; the device is business-ready |
| centralManagerDidDisconnectPeripheral:reason: | DHBleDisconnectReasonPasswordAuthFailed | Authentication failed; the SDK disconnects the device  |

Example:

```
//Configure the current account's four-digit password before connecting.
[DHbleCommand prepareAutoPassword:@"1234"];
[DHbleCentralManager connectDeviceWithModel:deviceModel];

//The device is business-ready only after authentication succeeds.
- (void)centralManagerDidFunctionMenu:(DeviceFuncV2Model *)deviceFuncModel
    peripheral:(DHPeripheralModel *)peripheral {
    NSLog(@"Device ready");
}
```

3.2.1.26.2 Modify the Device Password

```
+(void)modifyDevicePwd:(nullable NSString *)password completion:(void (^ _Nullable)(BOOL success))completion
```

Modify the device password after the device is connected and authenticated. `success == YES` means the device confirmed the change. For a normal unbind operation, first change the device password to `0000`; only clear the local binding and disconnect after the change succeeds.

Example:

```
[DHBleCommand modifyDevicePwd:@"0000" completion:^(BOOL success) {
    if (success) {
        [DHBleCentralManager setBindedStatus:NO];
        [DHBleCentralManager disconnectDevice];
    }
}];
```

3.2.1.27 Instant Screen Control

Check `isSupportScreenControl` in the device configuration table before using this feature.

Methods:

```
+(void)setScreenOn:(BOOL)isOn block:(void(^)(int code, id data))block
```

| Parameter | Type | Description                                                                  |
|-----------|------|------------------------------------------------------------------------------|
| isOn      | BOOL | <code>YES</code> : turn the screen on; <code>NO</code> : turn the screen off |

Example:

```
[DHBleCommand setScreenOn:YES block:^(int code, id _Nonnull data) {
    NSLog(@"set screen on, code=%d", code);
}];
```

3.2.2 Health data synchronization (real-time single measurement and all-day monitoring)

There are two ways to monitor health data: real-time single measurements and continuous 24-hour monitoring. Health data includes heart rate, blood oxygen, stress levels, HRV, and sleep, **but sleep is not monitored in real time.**

- (1) Real-time single detection: The app initiates a single detection on the device, and the results are returned immediately after the detection is complete.
- (2) Continuous monitoring: You can set the interval time, for example, 30 minutes or 60 minutes, and the device will perform measurements and save the data; **if the app is not synchronized, the device can store 3-6 days of data.**

3.2.2.1 Real-time monitoring - Start and stop device health data monitoring.

Start health data monitoring (heart rate, blood oxygen, HRV, stress, blood sugar);

After the test is completed, the device notifies the app via the `BluetoothNotificationHealthRingMeasureStateChange` notification;

The real-time test values are notified to the app via the `BluetoothNotificationHealthRingMeasureValueChange` notification;

### ⚠ Caution

Only one health detection type can be active at a time. You must wait for the current detection to complete (receive the completion callback) or manually stop it before starting a new detection type. Starting multiple types simultaneously will cause detection errors.

Method Description:

```
+(void)controlOpen:(NSInteger)type dataType:(NSInteger)dataType block:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type      | illustrate        | illustrate                                                                                                                                                                                      |
|-----------|-----------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dataType  | NSInteger | Health data types | Heart Rate: BLE_KEY_HEART_RATE<br>Blood Oxygen: BLEKEY_BLOOD_OXYGEN<br>HRV: BLE_KEY_HRV<br>Stress: BLE_KEY_STRESS<br>Blood Sugar: BLE_KEY_BLOOD_SUGAR<br>Blood Pressure: BLE_KEY_BLOOD_PRESSURE |
| type      | NSInteger | Start/Stop        | Start: 1<br>Stop: 0                                                                                                                                                                             |

Example of usage:

```
+ (void)controlOpen:(NSInteger)type dataType:(NSInteger)dataType block:(void(^)(int code, id
data))block
//启动心率测试
[DHBleCommand controlOpen:1 dataType:BLE_KEY_HEART_RATE block:^(int code, id _Nonnull data) {

}];

//关闭心率测试
[DHBleCommand controlOpen:0 dataType:BLE_KEY_HEART_RATE block:^(int code, id _Nonnull data) {

}];

// 监听测量中实时数值改变
[[NSNotificationCenter defaultCenter] addObserver:self
selector:@selector(updateRingMeasureValueChange:)
name:BluetoothNotificationHealthRingMeasureValueChange object:nil];

- (void)updateRingMeasureValueChange:(NSNotification *)ntf
{
    NSDictionary *userInfo = ntf.userInfo;
    NSInteger tDataValue = [userInfo[@"dataValue"] integerValue];
    NSInteger tDataType = [userInfo[@"dataType"] integerValue];

    NSLog(@"updateRingMeasureValueChange 0x%04X value: %zd", (unsigned int)tDataType, tDataValue);
    if (tDataType == BLE_KEY_APP_REAL_TIME_MUSLIM_COUNT){ //Muslim Count

    }
    else if (tDataType == BLE_KEY_APP_REAL_BLOOD_SUGAR_DATA){ //BloodSugar

    }
}
```

```

else if (tDataType == BLE_KEY_APP_REAL_TIME_HRV_DATA){ //HRV
}
else if (tDataType == BLE_KEY_APP_REAL_TIME_HR_DATA){ //HR Heart Rate
}
else if (tDataType == BLE_KEY_APP_REAL_TIME_BLOOD_OXYGEN_DATA){ //BloodOxygen
}
else if (tDataType == BLE_KEY_APP_REAL_TIME_STRESS_DATA){ //Stress
}
else if (tDataType == BLE_KEY_APP_REAL_TIME_BP_DATA){ //BloodPressure
    NSInteger sp = [tUserInfo[@"systolic"] integerValue]; //Systolic
    NSInteger dp = [tUserInfo[@"diastolic"] integerValue]; //Diastolic
    NSLog(@"BloodPressure sp=%zd dp=%zd", sp, dp);
}
}
}

```

### 3.2.2.2 Continuous monitoring - Set the interval for continuous monitoring of health data.

Set the monitoring interval for health data (heart rate, blood oxygen, HRV, stress, blood glucose) throughout the day, in minutes.

**Notes:** Currently, only the heart rate interval can be set to 30 minutes or 60 minutes; other parameters (blood oxygen, HRV, stress, blood glucose) can only be set to on or off. The start and end times are fixed to cover the entire day and cannot be modified.

#### 3.2.2.2.1 Heart rate detection settings and retrieval

The only interval options available for heart rate monitoring are 30 minutes and 60 minutes;

Method Description:

```
+(void)setHeartRateMode:(DHHeartRateModeSetModel *)model block:(void(^)(int code, id data))block
```

```
+(void)getHeartRateMode:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type                    |       |                                                                                                                                                                                                                                                        |
|-----------|-------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model     | DHHeartRateModeSetModel | class | isOpen: true (on)/false (off)<br>interval: interval time 30 or 60 minutes<br>startHour: 0 (fixed, cannot be modified)<br>startMin: 0 (fixed, cannot be modified)<br>endHour: 23 (fixed, cannot be modified)<br>endMin: 59 (fixed, cannot be modified); |

Example of usage:

```

// 1. Set HeartRate Monitor(设置心率监听)
DHHeartRateModeSetModel *tModeSetModel = [[DHHeartRateModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; //开始时间固定
tModeSetModel.startMinute = 00;

```

```
tModeSetModel.endHour = 23; //结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 30; //可设置30分钟或60分钟
[DHBleCommand setHeartRateMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setHeartRateMode OK");
    }
}];

//1. 获取心率监听
[DHBleCommand getHeartRateMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHHeartRateModeSetModel *model = data;
        NSLog(@"getHeartRateMode OK 开关 %d 检测周期 %d", model.isOpen, model.interval);
    }
}];
```

3.2.2.2.2 Blood oxygen monitoring settings and data retrieval

The interval for blood oxygen measurements can only be set to 60 minutes;

Method Description:

```
+(void)setBoMode:(DHBoModeSetModel *)model block:(void(^)(int code, id data))block
+(void)getBoMode:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type             |       |                                                                                                                                                                                                                                                      |
|-----------|------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model     | DHBoModeSetModel | Class | isOpen: true (on)/false (off)<br>interval: Fixed interval of 60 minutes<br>startHour: 0 (fixed, cannot be modified)<br>startMin: 0 (fixed, cannot be modified)<br>endHour: 23 (fixed, cannot be modified)<br>endMin: 59 (fixed, cannot be modified); |

Example of usage:

```
// 2. Set Blood oxygen Monitor(设置血氧监听)
DHBoModeSetModel *tModeSetModel = [[DHBoModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; //开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; //结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60; //固定不可设置
[DHBleCommand setBoMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setBoMode OK");
    }
}];

//2. 获取血氧监听
[DHBleCommand getBoMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHBoModeSetModel *model = data;
```

```
        NSLog(@"getBoMode OK 开关 %d 检测周期 %d", model.isOpen, model.interval);
    }
}];
```

3.2.2.2.3 Heart Rate Variability (HRV) Measurement Settings and Data Acquisition

The HRV interval can only be set to 60 minutes;

Method Description:

```
+(void)setHrvMode:(DHHrvModeSetModel *)model block:(void(^)(int code, id data))block

+(void)getHrvMode:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type              |       |                                                                                                                                                                                                                                                      |
|-----------|-------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model     | DHHrvModeSetModel | Class | isOpen: true (on)/false (off)<br>interval: Fixed interval of 60 minutes<br>startHour: 0 (fixed, cannot be modified)<br>startMin: 0 (fixed, cannot be modified)<br>endHour: 23 (fixed, cannot be modified)<br>endMin: 59 (fixed, cannot be modified); |

Example of usage:

```
// 3. Set HRV Monitor(设置HRV监听)
DHHrvModeSetModel *tModeSetModel = [[DHHrvModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; //开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; //结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60; //固定不可设置
[DHBleCommand setHrvMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setHrvMode OK");
    }
}];

//3. 获取HRV监听
[DHBleCommand getHrvMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHHrvModeSetModel *model = data;
        NSLog(@"getHrvMode OK 开关 %d", model.isOpen);
    }
}];
```

3.2.2.2.4 Stress detection settings and retrieval

The interval stress can only be set to 60 minutes;

Method Description:

```
+(void)setStressMode:(DHStressModeSetModel *)model block:(void(^)(int code, id data))block
```

```
+(void)getStressMode:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type                 |       |                                                                                                                                                                                                                                                      |
|-----------|----------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model     | DHStressModeSetModel | Class | isOpen: true (on)/false (off)<br>interval: Fixed interval of 60 minutes<br>startHour: 0 (fixed, cannot be modified)<br>startMin: 0 (fixed, cannot be modified)<br>endHour: 23 (fixed, cannot be modified)<br>endMin: 59 (fixed, cannot be modified); |

Example of usage:

```
// 4. 设置压力监听
DHStressModeSetModel *tModeSetModel = [[DHStressModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; // 开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; // 结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60; // 固定不可设置
[DHBleCommand setStressMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setStressMode OK");
    }
}];

// 4. 获取压力监听
[DHBleCommand getStressMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHStressModeSetModel *model = data;
        NSLog(@"getStressMode OK 开关 %d", model.isOpen);
    }
}];
```

3.2.2.2.5 Blood glucose monitoring settings and data retrieval

The interval between blood glucose measurements can only be set to 60 minutes;

Method Description:

```
+(void)setBloodSugarMode:(DHBloodSugarModeSetModel *)model block:(void(^)(int code, id data))block
```

```
+(void)getBloodSugarMode:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type                     |       |                                                                                                                                                                                                                                                      |
|-----------|--------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model     | DHBloodSugarModeSetModel | Class | isOpen: true (on)/false (off)<br>interval: Fixed interval of 60 minutes<br>startHour: 0 (fixed, cannot be modified)<br>startMin: 0 (fixed, cannot be modified)<br>endHour: 23 (fixed, cannot be modified)<br>endMin: 59 (fixed, cannot be modified); |

Example of usage:

```
// 5. 设置血糖监听
DHBloodSugarModeSetModel *tModeSetModel = [[DHBloodSugarModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; // 开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; // 结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60; // 固定不可设置
[DHBleCommand setBloodSugarMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setBloodSugarMode OK");
    }
}];

// 5. 获取血糖监听
[DHBleCommand getBloodSugarMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHBloodSugarModeSetModel *model = data;
        NSLog(@"getBloodSugarMode OK 开关 %d", model.isOpen);
    }
}];
```

3.2.2.2.6 Blood pressure monitoring settings and data retrieval

The interval for blood pressure measurements can only be set to 60 minutes;

Configuration table property: isDataTypeBloodPressure

Method Description:

```
+(void)setBpMode:(DHBpModeSetModel *)model block:(void(^)(int code, id data))block
+(void)getBpMode:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type             |       |                                                                                                                                                                                                                                                      |
|-----------|------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| model     | DHBpModeSetModel | Class | isOpen: true (on)/false (off)<br>interval: Fixed interval of 60 minutes<br>startHour: 0 (fixed, cannot be modified)<br>startMin: 0 (fixed, cannot be modified)<br>endHour: 23 (fixed, cannot be modified)<br>endMin: 59 (fixed, cannot be modified); |

Example of usage:

```
// 6. Set Blood Pressure Monitor(设置血压监听)
DHBpModeSetModel *tModeSetModel = [[DHBpModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; //开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; //结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60; //固定不可设置
[DHBleCommand setBpMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setBpMode OK");
    }
}];

//6. Get Blood Pressure Monitor
[DHBleCommand getBpMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHBpModeSetModel *model = data;
        NSLog(@"getBpMode OK isOpen %d", model.isOpen);
    }
}];
```

### 3.2.2.3 24/7 monitoring - Synchronized health history data

Synchronizing health history data will automatically sync the corresponding health data based on the device's capabilities.

`data` is an array containing multi-day data of a specific type. It will return data of that type sequentially; `data` will not contain data of multiple types.

```
[DHBleCommand startDataSyncing:^(int code, id data){
    NSLog(@"同步完成 %d", code);
} datablcok:^(int code, int progress, id _Nonnull data) {
    if (code == 0) {
        if ([data isKindOfClass:[NSArray class]]) {
            NSArray *array = data;
            for (id model in array) {
                if ([model isKindOfClass:[DHDailyStepModel class]]) { //Step
                    NSLog(@"同步有 计步数据");
                }
                else if ([model isKindOfClass:[DHDailySleepModel class]]) { //Sleep
                    NSLog(@"同步有 睡眠数据");
                }
            }
        }
    }
}];
```

```

        else if ([model isKindOfClass:[DHDailyHrModel class]]) { //HeartRate
            NSLog(@"同步有 心率数据");
        }
        else if ([model isKindOfClass:[DHDailyBoModel class]]) { //BO
            NSLog(@"同步有 血氧数据");
        }
        else if ([model isKindOfClass:[DHDailyHrvModel class]]) { //HRV
            NSLog(@"同步有 HRV数据");
        }
        else if ([model isKindOfClass:[DHDailyPressureModel class]]) { ///Stress
            NSLog(@"同步有 Stress数据");
        }
        else if ([model isKindOfClass:[DHDailyBloodSugarModel class]]) {
            NSLog(@"同步有 血糖数据");
        }
        else if ([model isKindOfClass:[DHDailyMuslimCountModel class]]) {
            NSLog(@"同步有 赞念数据");
        }
        else if ([model isKindOfClass:[DHDailyTempModel class]]) { ///Body
            NSLog(@"Body temperature data received");
        }
        else if ([model isKindOfClass:[DHDailyBpModel class]]) { ///Blood pressure
            NSLog(@"Blood pressure data received");
        }
    }
}
}
}
}];

```

### 3.2.2.4 24/7 Monitoring - Health Data Explanation

Each `dataBlock` call returns an array containing only one model type. Different health data types are not mixed in the same array.

**Time fields:** `timestamp`, `beginTime`, `endTime`, and item-level `timestamp` values in this section are Unix timestamps in seconds. Model time properties use `NSString`; item dictionary timestamps use `NSNumber`.

Data overview:

| Data             | Daily model             | Main fields in each <code>items</code> dictionary |
|------------------|-------------------------|---------------------------------------------------|
| Steps            | DHDailyStepModel        | timestamp, index, step, calorie, distance         |
| Sleep            | DHDailySleepModel       | status, value                                     |
| Heart rate       | DHDailyHrModel          | timestamp, value                                  |
| Blood pressure   | DHDailyBpModel          | timestamp, systolic, diastolic                    |
| Blood oxygen     | DHDailyBoModel          | timestamp, value                                  |
| Body temperature | DHDailyTempModel        | timestamp, value                                  |
| Stress           | DHDailyPressureModel    | timestamp, value                                  |
| Blood glucose    | DHDailyBloodSugarModel  | timestamp, value                                  |
| HRV              | DHDailyHrvModel         | timestamp, value                                  |
| Dhikr count      | DHDailyMuslimCountModel | timestamp, index, value                           |

Regular measurement data:

The heart rate, blood pressure, blood oxygen, body temperature, stress, blood glucose, and HRV daily models contain:

| Property  | Type                           | Description                             |
|-----------|--------------------------------|-----------------------------------------|
| timestamp | NSString                       | Unix timestamp for the date, in seconds |
| date      | NSString                       | Date in <code>yyyyMMdd</code> format    |
| items     | NSMutableArray<NSDictionary *> | Measurement details for the day         |

| Data             | Item fields                    | Unit or conversion                                                                     |
|------------------|--------------------------------|----------------------------------------------------------------------------------------|
| Heart rate       | timestamp, value               | bpm                                                                                    |
| Blood pressure   | timestamp, systolic, diastolic | systolic and diastolic pressure, in mmHg                                               |
| Blood oxygen     | timestamp, value               | %                                                                                      |
| Body temperature | timestamp, value               | Actual temperature = <code>value / 10.0</code> , in °C                                 |
| Stress           | timestamp, value               | Device stress value, no standard unit                                                  |
| Blood glucose    | timestamp, value               | On iOS, <code>value</code> is a numeric string; convert it before numerical processing |
| HRV              | timestamp, value               | ms                                                                                     |

**Important:** Blood pressure contains both systolic and diastolic values and must not be handled as single-value data.  
The blood glucose `value` is a string.

Step data — `DHDailyStepModel`:

| Property             | Type                           | Description                                                      |
|----------------------|--------------------------------|------------------------------------------------------------------|
| timestamp            | NSString                       | Unix timestamp for the date, in seconds                          |
| date                 | NSString                       | Date in <code>yyyyMMdd</code> format                             |
| step                 | NSInteger                      | Total steps for the day                                          |
| calorie              | NSInteger                      | Total calories for the day                                       |
| distance             | NSInteger                      | Total distance for the day, in meters                            |
| activityDataInterval | NSInteger                      | Detail interval in minutes; defaults to 60 when unconfigured     |
| items                | NSMutableArray<NSDictionary *> | Each item contains timestamp, index, step, calorie, and distance |

| Data    | progress in dataBlock | Returned content and daily totals                                                           |
|---------|-----------------------|---------------------------------------------------------------------------------------------|
| Today   | 1                     | Returns the current-day model; use the device-provided step, calorie, and distance totals   |
| History | 2                     | May return multiple daily models; each day's totals are accumulated from that day's details |

`activityDataInterval=60` means one item per hour; `10` means one item every 10 minutes. Use each `items.timestamp` value as the exact detail time.

Sleep data — `DHDailySleepModel`:

| Property  | Type                           | Description                                 |
|-----------|--------------------------------|---------------------------------------------|
| timestamp | NSString                       | Unix timestamp for the date, in seconds     |
| date      | NSString                       | Date in <code>yyyyMMdd</code> format        |
| duration  | NSInteger                      | Total sleep duration, in minutes            |
| beginTime | NSString                       | Sleep start time, Unix timestamp in seconds |
| endTime   | NSString                       | Wake-up time, Unix timestamp in seconds     |
| items     | NSMutableArray<NSDictionary *> | Each item contains status and value         |

In each sleep item, `value` is the stage duration in minutes. `status`: 0 awake, 1 light sleep, 2 deep sleep, 3 REM.

Dhikr count data — `DHDailyMuslimCountModel`:

| Property    | Type                           | Description                                    |
|-------------|--------------------------------|------------------------------------------------|
| timestamp   | NSString                       | Unix timestamp for the date, in seconds        |
| date        | NSString                       | Date in <code>yyyyMMdd</code> format           |
| muslimcount | NSInteger                      | Total count for the day                        |
| items       | NSMutableArray<NSDictionary *> | Each item contains timestamp, index, and value |

Dhikr details are returned hourly; `value` is the cumulative count for that hour.

### 3.2.3 OTA upgrade

#### Note

The OTA file must be provided by the manufacturer and confirmed for the current product. Before updating, follow [3.2.1.3 Get Device Information](#) to read `DHFWirmwareVersionModel.deviceModel` and compare it with the firmware target model supplied by the manufacturer. Start the update only when they match. Abort when the model is empty or different to prevent an incompatible firmware file from making the device unusable.

#### 3.2.3.1 Get Available Firmware

Use the following endpoint to query the available firmware list for a device model:

```
GET https://ruiwo168.com/api/device/getOtaListByModel?model=<deviceModel>
```

The `model` parameter corresponds to `DHFWirmwareVersionModel.deviceModel` returned by `getFirmwareVersion`. Read the device firmware information first and use the actual model reported by the device.

```
[DHBLECommand getFirmwareVersion:^(int code, id _Nonnull data) {
    if (code != 0 || ![data isKindOfClass:[DHFWirmwareVersionModel class]]) {
        return;
    }
    DHFWirmwareVersionModel *version = data;
    if (version.deviceModel.length == 0) {
        return;
    }
    NSString *model = [version.deviceModel
stringByAddingPercentEncodingWithAllowedCharacters:NSCharacterSet.URLQueryAllowedCharacterSet];
    NSString *url = [NSString stringWithFormat:@"https://ruiwo168.com/api/device/getOtaListByModel?model=%@", model];
    NSLog(@"query firmware: %@, currentVersion=%@", url, version.firmwareVersion);
    // Request this URL using the app's existing networking component.
}];
```

Example response:

```

{
  "code": 0,
  "msg": "操作成功",
  "data": [
    {
      "deviceModel": "DEVICE_MODEL",
      "toVersion": "X.Y.Z",
      "size": 123456,
      "downloadUrl": "https://example.com/path/firmware.bin"
    }
  ]
}

```

| Field       | Type   | Description                                                                                   |
|-------------|--------|-----------------------------------------------------------------------------------------------|
| deviceModel | String | Target device model; it must exactly match <code>DHFWirmwareVersionModel.deviceModel</code>   |
| toVersion   | String | Target firmware version, used in production to determine whether a newer version is available |
| size        | Int    | Firmware file size in bytes                                                                   |
| downloadUrl | String | Firmware download URL                                                                         |

For production releases, compare the current `firmwareVersion` with the target `toVersion` numerically by each `x.y.z` segment and normally prompt only for a newer version. Do not compare versions as plain strings. For testing, the same version or a downgrade may be installed after confirming that the firmware is valid.

Verify that `deviceModel` matches before downloading and again before upgrading. After downloading the firmware, load it as `NSData` and pass it to `ringOtaWithFileData`. When hosting firmware on your own server, maintain the mapping between device models, versions, and firmware files.

### 3.2.3.2 Perform OTA Upgrade

Pre-update validation:

| Data                  | Source                                               | Purpose                                                          |
|-----------------------|------------------------------------------------------|------------------------------------------------------------------|
| deviceModel           | <code>DHFWirmwareVersionModel.deviceModel</code>     | Current device model and unique identifier for the product model |
| Firmware target model | Supplied by the manufacturer with the firmware file  | Must exactly match deviceModel                                   |
| firmwareVersion       | <code>DHFWirmwareVersionModel.firmwareVersion</code> | Can be used to decide whether an update is required              |

Method Description:

```

+(void)ringOtaWithFileData:(NSData *)fileData block:(void(^)(int code, CGFloat progress, id data))block

```

Parameter Description:

| Parameter | Type   | Description                                                       |
|-----------|--------|-------------------------------------------------------------------|
| fileData  | NSData | Firmware file data                                                |
| block     | Block  | Progress and result callback, code=0 in progress, progress is 0-1 |

Example of usage:

```
[DHBleCommand getFirmwareVersion:^(int code, id _Nonnull data) {
    if (code != 0 || ![data isKindOfClass:[DHFirmwareVersionModel class]]) {
        return;
    }

    DHFirmwareVersionModel *version = data;
    NSString *firmwareTargetModel = @"target model supplied by manufacturer";
    if (version.deviceModel.length == 0 ||
        ![version.deviceModel isEqualToString:firmwareTargetModel]) {
        NSLog(@"Device model mismatch; OTA is not allowed");
        return;
    }

    NSString *filePath = @""; // Firmware file path supplied by manufacturer
    NSData *fileData = [NSData dataWithContentsOfFile:filePath];
    if (fileData.length == 0) {
        return;
    }
    [DHBleCommand ringOtaWithData:fileData block:^(int otaCode, CGFloat progress, id _Nonnull
otaData) {
        NSLog(@"OTA code %d progress %.2f", otaCode, progress);
    }];
}];
```

### 3.2.4 Exercise more


**Caution**

The multi-sport configuration table property is `isSupportWorkout3`. After enabling multi-sport mode, the device will enter exercise mode. Neither disconnecting the app nor closing it will stop the activity; it can only be stopped manually through the app or the device itself. Therefore, for devices with multi-sport functionality, please check the status after connecting to determine if it is currently in exercise mode, as this may affect the use of other functions.

**The exercise duration must exceed 2 minutes for the device to save the workout data.**

#### 3.2.4.1 Get the device's multiple motion states.

Check if the device is currently engaged in multiple activities; only start a new activity if it's not currently engaged in any activity.

Method Description:

```
+(void)getControlSportWithRing:(void(^)(int code, id data))block
```

Parameter Description:

| WorkoutControlType | Type |    |          |
|--------------------|------|----|----------|
| Workout_Begin      | Int  | 整形 | Begin    |
| Workout_Continue   | Int  | 整形 | Continue |
| Workout_Pause      | Int  | 整形 | Pause    |
| Workout_Finish     | Int  | 整形 | Finish   |

Example of usage:

```
[DHBleCommand getControlSportWithRing:^(int code, id _Nonnull data) {
    // @{@"keySportType":@"(tSportType), @"keyControlType":@"(tControlType)}
    if (code == 0 && [data isKindOfClass:[NSDictionary class]]){
        NSDictionary *tDic = data;
        NSInteger tSportType = [tDic[@"keySportType"] integerValue];
        WorkoutControlType tControlType = [tDic[@"keyControlType"] integerValue];
    }
}];
```

### 3.2.4.2 The control device enters multi-motion mode.

The control device enters multi-motion mode and starts the motion.

Changes in exercise data are obtained by receiving the BluetoothNotificationRingRuningData notification.

Method Description:

```
+(void)controlSportWithRing:(DHSportControlModel *)model block:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter           | Type  |  |                                                                                 |
|---------------------|-------|--|---------------------------------------------------------------------------------|
| DHSportControlModel | Class |  | controlType: refer to WorkoutControlType<br>sportType: refer to BleActivityMode |

Explanation of the data returned in the notification of changes in exercise data:

| Parameter        | Type |    |                                                                 |
|------------------|------|----|-----------------------------------------------------------------|
| ActivityTime     | Int  | 整形 | Duration of exercise, in seconds (s);                           |
| ActivitySteps    | Int  | 整形 | Steps taken during exercise                                     |
| ActivityDistance | Int  | 整形 | Distance is generated during movement, measured in meters (m);  |
| ActivityCalorie  | Int  | 整形 | Heat is generated during exercise, measured in calories (cal);  |
| ActivityHr       | Int  | 整形 | Dynamic heart rate during exercise                              |
| ActivityDataType | Int  | 整形 | The source type, which is also returned by setRingEnterWorkOut. |

Example of usage:

```

DHSportControlModel *model = [[DHSportControlModel alloc] init];
model.controlType = Workout_Begin; //开始
model.sportType = tbleActivityMode;
[DHBleCommand controlSportWithRing:model block:^(int code, id _Nonnull data) {
    if (code == 0){
        WorkoutRunningController *runningC = [[WorkoutRunningController alloc]
initWithNibName:@"WorkoutRunningController" bundle:nil];
        runningC.bleActivityMode = tbleActivityMode;
        runningC.controllType = Workout_Begin; //开始
        runningC.modalPresentationStyle = UIModalPresentationOverFullScreen;
        [weakSelf presentViewController:runningC animated:YES completion:^(
    }];
}
}];

[[NSNotificationCenter defaultCenter] addObserver:self selector:@selector(ringRuningDataUdpate:)
name:BluetoothNotificationRingRuningData object:nil];

- (void)ringRuningDataUdpate:(NSNotification *)ntf
{
    NSDictionary *tUserInfo = ntf.userInfo;
    if (tUserInfo.count > 0){
        NSInteger duration = [tUserInfo[@"ActivityTime"] integerValue];
        NSInteger totalStep = [tUserInfo[@"ActivitySteps"] integerValue];
        NSInteger activityDistance = [tUserInfo[@"ActivityDistance"] integerValue];
        NSInteger activityCalorie = [tUserInfo[@"ActivityCalorie"] integerValue];
        NSInteger activityHr = [tUserInfo[@"ActivityHr"] integerValue];
        NSInteger tActivityDataType = [tUserInfo[@"ActivityDataType"] integerValue];

        NSLog(@"ringRuningDataUdpate tActivityDataType %04X", (UInt32)tActivityDataType);
        NSInteger hours = duration / 3600;
        NSInteger minutes = (duration % 3600) / 60;
        NSInteger seconds = duration % 60;
        self.workoutTimeLb.text = [NSString stringWithFormat:@"Time: %02ld:%02ld:%02ld", (long)hours,
(long)minutes, (long)seconds];
        self.workoutStepLb.text = [NSString stringWithFormat:@"Steps: %ld", totalStep];
        self.workoutDistanceLb.text = [NSString stringWithFormat:@"Distance: %.2f Km",
floor(activityDistance/1000.0*100)/100];
        self.workoutCaloriesLb.text = [NSString stringWithFormat:@"Calories: %.1f KCal",
floor(activityCalorie/1000.0*10)/10];
        self.workoutHeartLb.text = [NSString stringWithFormat:@"HeartRate: %ld bpm", activityHr];
    }
}
}

```

The corresponding names for BleActivityMode can be found in the example Demo string definitions:

|                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> DHBleSDKDemo DHBleSDKDemo Base Resource Localizable Localizable (English) Localizable (Chinese, Simplified) AppDelegate AppDelegate AppDelegate AppDelegate AppDelegate </div> | <pre> 8 // 9 "str_jl_Running" = "Running"; 10 "str_jl_Treadmill" = "Treadmill"; 11 "str_jl_Outdoorrunning" = "Outdoor running"; 12 "str_jl_Cycling" = "Riding"; 13 "str_jl_Swim" = "Swim"; 14 "str_jl_Walking" = "Walking"; 15 "str_jl_Climbing" = "Mountain climbing"; 16 "str_jl_Yoga" = "Yoga"; 17 "str_jl_Spinning" = "spinning bike"; 18 "str_jl_Basketball" = "basketball"; 19 "str_jl_Football" = "football"; </pre> | <pre> 446 BLE_ACTIVITY_START_INDEX = 7, // 枚举值开始标记 447 BLE_ACTIVITY_RUNNING = 7, // 跑步 448 BLE_ACTIVITY_INDOOR = 8, // 跑步机 449 BLE_ACTIVITY_OUTDOOR = 9, // 户外跑步运动 450 BLE_ACTIVITY_CYCLING = 10, // 骑行 451 BLE_ACTIVITY_SWIMMING = 11, // 游泳 452 BLE_ACTIVITY_WALKING = 12, // 步行, 健走 453 BLE_ACTIVITY_CLIMBING = 13, // 爬山 454 BLE_ACTIVITY_YOGA = 14, // 瑜伽 455 BLE_ACTIVITY_SPINNING = 15, // 动感单车 456 BLE_ACTIVITY_BASKETBALL = 16, // 篮球 457 BLE_ACTIVITY_FOOTBALL = 17, // 足球 </pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

3.2.4.3 Control enabling/disabling real-time notifications of motion data from the device.

控制开启/关闭设备实时通知运动数据;

运动中数据变化通过接收 BluetoothNotificationRingRuningData 通知获取,有时app关闭与进入后台,可告诉设备停止通知数据.

Method Description:

```
+(void)setRingEnterWorkOut:(UInt8)isEnter block:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type |  |                                                                                       |
|-----------|------|--|---------------------------------------------------------------------------------------|
| isEnter   | Int  |  | 1: Enable notification of exercise data;<br>0: Disable notification of exercise data; |

Example of usage:

```
//退出运动界面
[DHBleCommand setRingEnterWorkOut:0 block:^(int code, id _Nonnull data) {

}];
```

3.2.4.4 Obtain multi-sport data reports.

Method Description:

```
+(void)startRingWorkout3Syncing:(void(^)(int code, id data))block dataBlock:(void(^)(int code, int progress, id data))dataBlock
```

Return data DHDailySportModel parameter description:

| DHDailySportModel<br>类 | Type     |  |                                                                                                                                                                                                                              |
|------------------------|----------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timestamp              | NSString |  | Exercise start timestamp                                                                                                                                                                                                     |
| date                   | NSString |  | Date-yyyyMMdd                                                                                                                                                                                                                |
| viewType               | Int      |  | Current exercise types include: steps, cadence, pace, and distance:<br>With cadence: viewTypeHaveStepFaq:<br>Without step count: viewTypeNoStepNum:<br>With pace: viewTypeHavePace:<br>Without distance: viewTypeNoDistance: |
| heartRateItems         | Array    |  | The current list of heart rates generated during exercise, saved every minute;                                                                                                                                               |
| pacePerKmlItems        | Array    |  | Pace per kilometer list, unit: seconds/km; empty if device does not support                                                                                                                                                  |
| .....                  |          |  | Other attributes can be found in the class documentation.                                                                                                                                                                    |

Example of usage:

```
[DHBluetooth startRingWorkout3Syncing:^(int code, id _Nonnull data) {
    NSLog(@"startRingWorkout3Syncing 同步完成 code %d", code);
} dataBlock:^(int code, int progress, id _Nonnull data) {
    if (code == 0) {
        if ([data isKindOfClass:[NSArray class]]){
            NSArray *array = data;
            for (id model in array) {
                if ([model isKindOfClass:[DHDailySportModel class]]) {
                    //保存数据库或其它操作

                }
            }

            NSLog(@"startRingWorkout3Syncing data %zd", array.count);

        }
    }
}];
```

### 5.2.5 Sensor Raw Data

This section covers two different data retrieval methods:

| Data                        | Retrieval method  | Description                                                                                       |
|-----------------------------|-------------------|---------------------------------------------------------------------------------------------------|
| PPG/ACC/PPG Red/IR raw data | History retrieval | The APP starts and stops collection, then actively synchronizes the stored data                   |
| Sleep state data            | Real-time push    | The device automatically pushes data during sleep; the APP only needs to observe the notification |

#### Important

PPG/ACC/PPG Red/IR raw data does not support real-time push and is available only through history retrieval. Sleep state data uses only real-time push and is not retrieved through the historical raw data API.

The current historical raw data sampling rate can reach up to 100 Hz, with a maximum test duration of approximately one minute. Individual samples do not contain timestamps, so the absolute time of each sample cannot be reconstructed.

Function table properties: `isSupportSensorRawPPG` (PPG), `isSupportSensorRawACC` (ACC), `isSupportSensorRawPPGRed` (PPG Red), `isSupportSensorRawIR` (IR), and `isSupportSensorRawSleep` (sleep real-time data).

Valid `sensorType` combinations for PPG/ACC/PPG Red/IR historical collection:

| Value | Meaning              | Description                          |
|-------|----------------------|--------------------------------------|
| 1     | ACC                  | ACC only                             |
| 2     | PPG Green            | PPG Green only                       |
| 3     | PPG Green + ACC      | PPG Green and ACC simultaneously     |
| 4     | PPG Red              | PPG Red only                         |
| 5     | PPG Red + ACC        | PPG Red and ACC simultaneously       |
| 10    | PPG Green + IR       | PPG Green and IR simultaneously      |
| 11    | PPG Green + ACC + IR | PPG Green, ACC and IR simultaneously |
| 12    | PPG Red + IR         | PPG Red and IR simultaneously        |
| 13    | PPG Red + ACC + IR   | PPG Red, ACC and IR simultaneously   |

**Rules: PPG Green and PPG Red cannot coexist; IR cannot start alone, must be combined with PPG Green or PPG Red.**

**Note:** The control API's `sensorType` is a sensor bitmask, while the returned dictionary's `sensorType` is a data type. They use different numbering. For example, control `sensorType=1` starts ACC, while historical data `sensorType=1` means PPG. Control `sensorType=5` starts PPG Red + ACC, while sleep real-time data `sensorType=5` means a sleep state.

#### 5.2.5.0 PPG Timed Monitoring

PPG timed monitoring setting, similar to heart rate/HRV timed monitoring;

Configuration table property: `isSupportPPGMonitoring`

Method Description:

```
+(void)setPPGMode:(DHHrvModeSetModel *)model block:(void(^)(int code, id data))block
```

```
+(void)getPPGMode:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter | Type              | Description | Value                                                                                                                                               |
|-----------|-------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| model     | DHHrvModeSetModel | Class       | isOpen: true on/false off<br>interval: default 30 minutes<br>startHour: 0 fixed<br>startMinute: 0 fixed<br>endHour: 23 fixed<br>endMinute: 59 fixed |

Example of usage:

```
//Set PPG monitoring
DHHrvModeSetModel *tModeSetModel = [[DHHrvModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00;
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23;
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60;
[DHBleCommand setPPGMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){ NSLog(@"setPPGMode OK"); }
}];

//Get PPG monitoring
[DHBleCommand getPPGMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHHrvModeSetModel *model = data;
        NSLog(@"getPPGMode OK isOpen %d", model.isOpen);
    }
}];
```

### 5.2.5.1 Start and Stop Sensor Raw Data

This API controls only PPG/ACC/PPG Red/IR historical raw data collection. It is not required for sleep real-time data.

block callback with code==0 indicates start/stop success;

The device may also stop the sensor actively, notified via `BluetoothNotificationHealthRingSenorStopChange`.

Method Description:

```
+(void)ringControlSensorRaw:(UInt8)outputType type:(UInt8)sensorType block:(void(^)(int code, id data))block
```

Parameter Description:

| Parameter  | Type  | Description           | Value                                           |
|------------|-------|-----------------------|-------------------------------------------------|
| outputType | UInt8 | Output control type   | 1: Start Sensor output<br>2: Stop Sensor output |
| sensorType | UInt8 | Sensor type (bitmask) | See valid combinations table above              |

Example of usage:

```
//Listen for device-initiated sensor stop
[[NSNotificationCenter defaultCenter] addObserver:self selector:@selector(sensorStopChange:)
name:BluetoothNotificationHealthRingSenorStopChange object:nil];

- (void)sensorStopChange:(NSNotification *)ntf {
    NSLog(@"Device stopped sensor actively");
}

//Start PPG+ACC raw data output (sensorType=3)
[DHBleCommand ringControlSensorRaw:1 type:3 block:^(int code, id data) {}];

//Stop PPG+ACC raw data output
[DHBleCommand ringControlSensorRaw:2 type:3 block:^(int code, id data) {}];

// Remove the observer when the page is destroyed or no longer needs the event.
[[NSNotificationCenter defaultCenter] removeObserver:self
                                name:BluetoothNotificationHealthRingSenorStopChange
                                object:nil];
```

5.2.5.2 Historical Raw Data Retrieval

PPG/ACC/PPG Red/IR raw data is available only through history retrieval. The device collects and stores the data first, and the APP later actively synchronizes it using `ringGetHistorySensorRaw`.  
Data is returned via `dataBlock` callback. `block` with `code==0` indicates sync is complete.

Method Description:

```
+(void)ringGetHistorySensorRaw:(void(^)(int code, id data))block dataBlock:(void(^)(int code, int progress, id data))dataBlock
```

`dataBlock` returns `NSArray<NSDictionary *>`; each dictionary represents one sensor data packet:

| Field      | Type                    | Description                                                                                                                            |
|------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| sensorType | NSNumber                | Data type: 1=PPG, 2=ACC, 3=PPG Red, 4=IR                                                                                               |
| sequence   | NSNumber                | Packet sequence starting from 1 and incrementing for every returned packet; all simultaneously enabled sensors share the same sequence |
| count      | NSNumber                | Number of samples in the current packet                                                                                                |
| ppgData    | NSArray<NSNumber *>     | PPG samples, each int32; present only when sensorType=1                                                                                |
| accData    | NSArray<NSDictionary *> | ACC samples, each containing x, y, and z int16 values; present only when sensorType=2                                                  |
| ppgRedData | NSArray<NSNumber *>     | PPG Red samples, each int32; present only when sensorType=3                                                                            |
| irData     | NSArray<NSNumber *>     | IR samples, each int32; present only when sensorType=4                                                                                 |

`dataBlock` is called only once with the complete result array, and `progress` is always 100. Raw samples do not contain individual timestamps.

Example of usage:

```
[DHBleCommand ringGetHistorySensorRaw:^(int code, id _Nonnull data) {
    NSLog(@"Sensor History sync finished code %d", code);
} dataBlock:^(int code, int progress, id _Nonnull data) {
    if (code == 0 && data) {
        NSArray *resultArray = data;
        for (NSDictionary *info in resultArray) {
            NSLog(@"sensorType=%@ seq=%@ count=%@", info[@"sensorType"], info[@"sequence"],
info[@"count"]);
        }
    }
}];
```

5.2.5.3 Sleep State Real-Time Push

Sleep state data supports only real-time push. Do not call `ringControlSensorRaw` to start or stop it. When supported by the device, sleep state data is automatically pushed during sleep.

Function table property: `isSupportSensorRawSleep`.

Receive it through the `BluetoothNotificationSensorRawData` notification. The notification `userInfo` contains:

| Field      | Type                    | Description                                             |
|------------|-------------------------|---------------------------------------------------------|
| sensorType | NSNumber                | Fixed at 5, indicating sleep state data                 |
| sleepData  | NSArray<NSDictionary *> | Sleep state list; each item contains timestamp and mode |

`sleepData` item:

| Field     | Type     | Description               |
|-----------|----------|---------------------------|
| timestamp | NSNumber | Unix timestamp in seconds |
| mode      | NSNumber | Sleep mode                |

Sleep modes:

| Value | Description |
|-------|-------------|
| 17    | Sleep start |
| 34    | Sleep end   |
| 1     | Deep sleep  |
| 2     | Light sleep |
| 3     | Awake       |
| 4     | REM         |

Example:

```
// Register when the page is initialized.
[[NSNotificationCenter defaultCenter] addObserver:self
```

```
        selector:@selector(sensorRawDataUpdate:)
        name:BluetoothNotificationSensorRawData
        object:nil];

- (void)sensorRawDataUpdate:(NSNotification *)notification {
    NSDictionary *userInfo = notification.userInfo;
    if ([userInfo[@"sensorType"] integerValue] != 5) {
        return;
    }
    NSArray<NSDictionary *> *sleepData = userInfo[@"sleepData"];
    for (NSDictionary *item in sleepData) {
        NSLog(@"sleep timestamp=%@ mode=%@", item[@"timestamp"], item[@"mode"]);
    }
}

// Remove when the page is destroyed or no longer needs the data.
[[NSNotificationCenter defaultCenter] addObserver:self
        name:BluetoothNotificationSensorRawData
        object:nil];
```

## SDK Revision History

---

### V2.0.0\_20260817 (2026.08.17)

- Added instant screen control (3.2.1.27).
- Added the available-firmware endpoint and OTA device-model/version validation guidance (3.2.3.1).

### V2.0.0\_20260724 (2026.07.24)

- Added support for step-detail intervals.

### V2.0.0\_20260706 (2026.07.06)

- Added external `CBCentralManager` integration (3.1.8), allowing the customer App to scan devices with its unified scanner and let the SDK connect with the same Central.

## 联系方式 / 技术支持

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